Is the *Blue Whiting* suffering from overfishing?

Introduction

Micromesistius poutassou (Figure 1) is a mesopelagic gadid that inhabits both the Mediterranean and the North Atlantic. It is considered that they are two isolated populations. The most common size ranges between 15 and 30 cm when they reach 4-5 years but they can live more than 10 years, even 20 in captivity (García, J. M. 2022).

M. poutassou is distributed between 150 and 3000 m of depth with the largest schools of fish between 300-400 m, although juveniles are usually found at shallower depths than adults. Individuals make vertical migrations throughout the day to feed or protect themselves and north-south migrations associated with reproduction (García, J. M. 2022).

Regarding the reproductive success of the species, they reach their first maturity when they are between 0-4 years old depending on the geographical area they inhabit and when their size is around 15-18 cm (L50, length at which 50% of the fish in a population reach sexual maturity). They reproduce between December and March (Mir-Arguimbau *et al*. 2020).

M. poutassou is a commercial species, that is, it is caught or farmed for sale, and therefore has a minimum size of 15 cm in the case of the Mediterranean, but there is no minimum size for the Atlantic (Ministry of Agriculture, Fisheries and Food).

Figure 1: *Micromesistius poutassou.*

Overfishing is a serious problem and therefore it is necessary to analyze the populations of commercial species such as the mare to find out if more individuals are being extracted than what is marked by the Maximum Sustainable Yield (MSY). Overfishing generates impacts on both marine ecosystems and economic or social. Eliminating this practice is essential to guarantee the resilience of the populations by making them recover and remain healthy (Ministry of Agriculture, Fisheries and Food).

There are indicators of overfishing such as the percentage of mature fish caught (it is expected to be 100%), the percentage of individuals that meet the minimum size (it is expected to be 100%) or the age of the individuals (this being the appropriate one for each species) (Froese, R. 2004).

Overfishing can generate problems such as precocious maturity, which is the ability of individuals to mature sexually at a much earlier age than is typical for the species. This increase in the speed of maturation leads to

a reduction in body growth. It is a mechanism that fish acquire to cope with sustained fishing pressure over time (Ministry of Agriculture, Fisheries and Food).

Objectives

By studying the populations of *M. poutassou* in the Mediterranean and the Atlantic, we want to find out if any of them are suffering from fishing pressure and, if so, give recommendations for sustainable management and fishing of the populations.

Methodology

To carry out the analysis, 2 batches of Micromesistius poutassou individuals were obtained, caught in 2 different areas: Mediterranean and Atlantic. The sample size was 74 and 78 respectively.

The following parameters were measured for each individual:

- Weight, measured with a scale with a precision of 1 gram.
- Total length, measured with an ichthyometer with a precision of 0.5mm.
- Maturity, based on visual analysis of the gonads (mature or immature).
- Sex, based on visual analysis of the gonads (male, female or immature).
- Age, by counting the slow and fast growth rings of the otoliths extracted from each individual.

Both statistical analysis and graphic representation with Visual studio code. The libraries used were pandas, seaborn, scipy and matplotlib. For the statistical tests, a t-test and a linear regression model were performed, establishing a 95% confidence interval.

Results

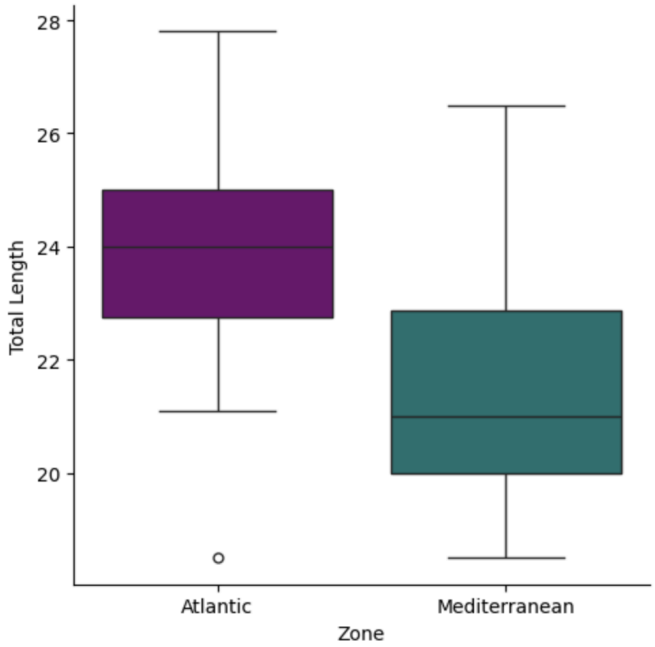

Figure 2: Total length of individuals by zone

Performing a t-test of the length of the two areas gives us a p-value < 0.05 and therefore we can say that there are significant differences.
In Figure 2 we see that the individuals from the Atlantic (purple) are larger than those from the Mediterranean (green). The average size of the individuals from the Atlantic measured is 24.04 cm while those from the Mediterranean are 21.46 cm. These data coincide with the typical size of *Micromesistius poutassou* (between 15-30 cm).

Taking into account the minimum catch size of the blue whitings established in the Mediterranean (15 cm), we can affirm that this measure is being carried out since the individuals, both from the Mediterranean and the Atlantic, exceed this measure.

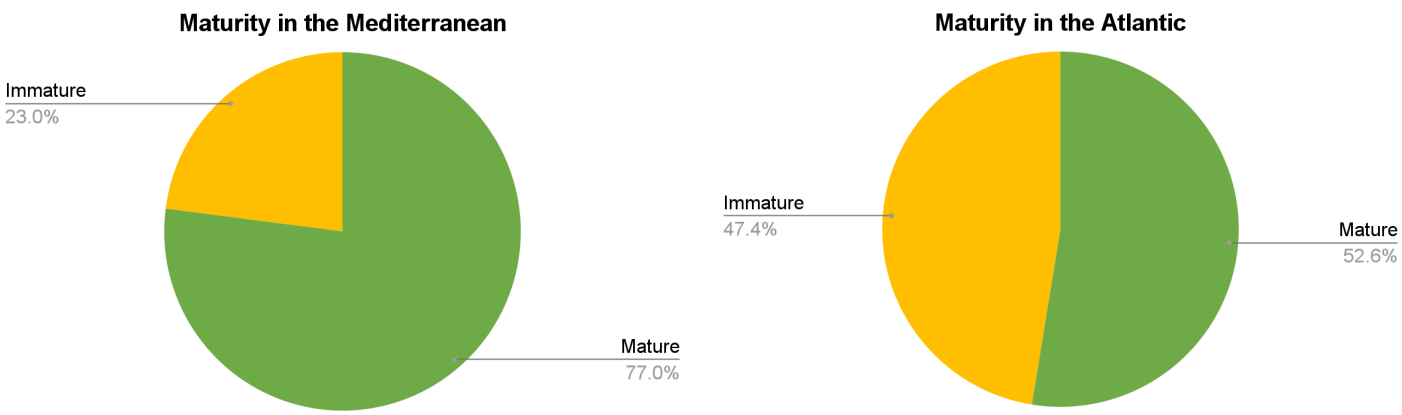

Figure 3: Percentage of mature and immature individuals by zone

Calculating the percentage of mature and immature individuals in both the Mediterranean and the Atlantic (Figure 3) we see that the Mediterranean fish are mostly mature (77% of the total) while in the Atlantic only 52% of the individuals measured are mature.

If we take into account one of the indicators of overfishing such as the percentage of mature individuals caught we could state that the Atlantic fishery is being carried out above the MSY levels, while in the Mediterranean the fishery would be closer to the MSY. However, it is seen that the average size of the Atlantic individuals is almost 3 cm larger.

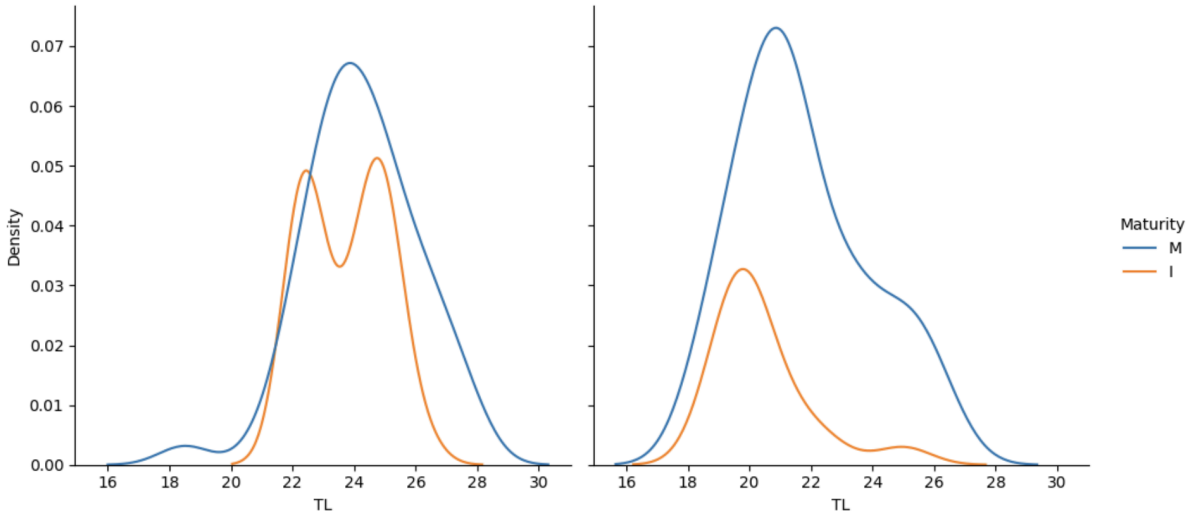

Figure 4: Density of individuals from the Atlantic (left) and the Mediterranean (right) and their total length by zone and maturity

Analyzing the density of individuals according to their total length (separated into mature and immature) in the Atlantic and the Mediterranean, we observe how the results of the previous graphs (Figures 2 and 3) are corroborated. In addition, we see how in the Atlantic, mature and immature individuals have a similar size (they follow a similar distribution) while in the case of the Mediterranean, immature individuals are smaller than mature ones (about 2 cm on average). This makes us suspect that in the Mediterranean, early maturity is occurring instead, since when individuals reach L50 (15-18 cm) they mature quickly to be able to reproduce, as a result of suffering fishing pressure. On the other hand, in the Atlantic, maturity is much more gradual, the juveniles do not have prey to reproduce and therefore spend more time growing.

To solve this, the minimum size in the Mediterranean could be increased from 15 to 21 cm, thus allowing the number of immatures to be reduced by 50% (Figure 4). Another more effective measure would be to implement a fishing ban during the reproductive season, from December to March. (Mir-Arguimbau *et al*. 2020) in the Mediterranean to allow time for individuals to mature more gradually and not so suddenly.

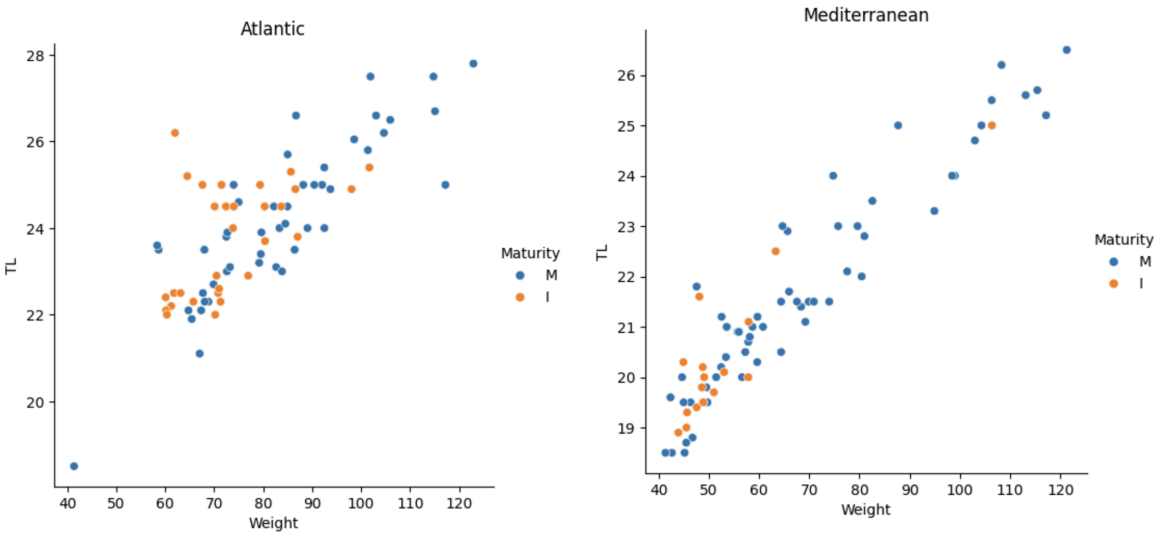

Figure 5. Scatter plot relating total length and weight

The figures show the maturation trend of *Micromesistius poutassou* (Figure 5). In the Mediterranean, immature fish are concentrated at the beginning of the graph, while in the Atlantic they overlap with the mature ones. Once again, it is seen that in the Atlantic they have more time to develop. There is also a difference in size, the Atlantic individuals are all over 22cm.

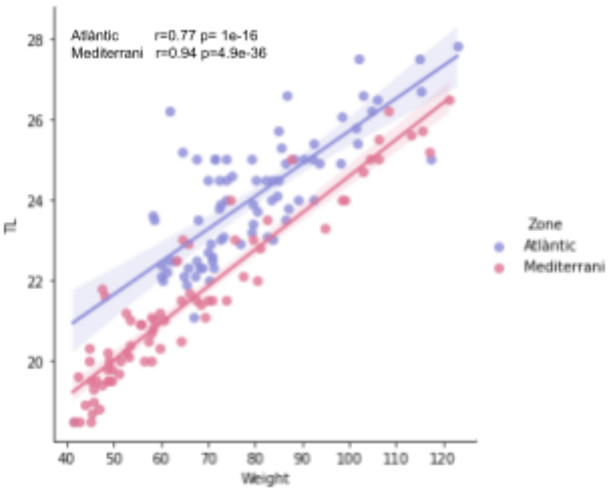

Figure 6. Linear regression model between weight and length by zone

Figure 6 shows the relationship between weight and length with respect to the two zones. The 2 straight lines show a steep slope that is confirmed with an r> 0.5 and a p-value <0.05. Normally, animals have a growth curve characterized by a linear phase, where they are growing until they reach a saturation phase where the changes in weight and length are smaller. Therefore, figure 6 indicates that the individuals are in

an initial phase of growth. If the r values are compared, it is found that the Mediterranean is closer to 1, which means that they reach the saturation point before the ones in the Atlantic. Due to fishing pressure they have less time to develop and end up being smaller and weighing less.

For this reason, we would expect there to be a ratio of more or less 50/50 between mature and immature individuals since they are growing. However, it has been previously seen that in the Mediterranean the majority are mature individuals, a higher percentage than in the Atlantic (Fig 3). This supports the idea that these fish suffer from early maturity.

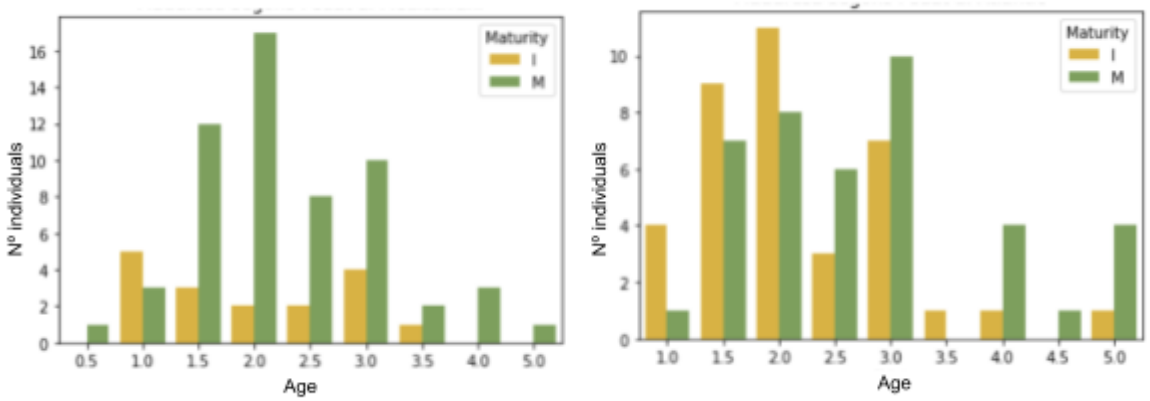

Figura 7. Number of individuals by age and maturity in Atlantic (left) and Mediterranean (right)

The graphs in figure 7, reaffirm the dynamics of the growth of the Atlantic. It is seen that there are more immatures at all ages while in the Mediterranean the green color predominates, that is, mature individuals,

Therefore, the management strategy cannot be the same for the two areas. In the Atlantic it would not be useful to establish a minimum size since at different lengths (Fig 5) and ages (Fig 7) we find both mature and immature. On the other hand, it is recommended to fish at greater depths, since it has been seen that the proportion of matures increases (García, J. M., 2022) and again, to establish a temporary closure during the reproduction season.

Conclusions

In the Mediterranean there is fishing pressure that leads to early maturity and therefore:

- The minimum size must be increased to 21cm.
- A closed period must be established during the reproductive season (December-March).

In the Atlantic there is fishing pressure that leads to a high catch of immatures, possible improvements would be:

- Establish a closed period during the reproductive season (January-March).
- Find strategies to fish at greater depths.

Finally, it is necessary to monitor and monitor these measures over a minimum period of 2 years to analyze maturity, along with more exhaustive research and study of the two populations.

Bibliography

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